CARD 1 OF 8 BULB COLORS **QUESTION**

A glass-bulb sprinkler has a **red** bulb. What temperature class is it? Then name the class for every other bulb color.

FSM · Fire System Mastery firesystemmastery.com

CARD 2 OF 8 FRAME COLORS **QUESTION**

A **solder-link** sprinkler has its frame arms painted **red**. What class is it?

FSM · Fire System Mastery firesystemmastery.com

CARD 3 OF 8 BULB HEADS **QUESTION**

On a **glass-bulb** sprinkler, what has to carry the color code: the frame arms or the liquid in the bulb?

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CARD 4 OF 8 NO COLOR NEEDED **QUESTION**

Which sprinklers **don't** have to show a temperature color at all? And how can a **corrosion-resistant** head show its color?

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CARD 2 OF 8 FRAME COLORS **ANSWER**

Extra high. Frame color codes run differently from bulb colors:
Uncolored or black: ordinary
White: intermediate
Blue: high
Red: extra high
Green: very extra high
Orange: ultra high

NFPA 13 (2019) · Ch. 7

Trap: red and orange **bulbs** are the coolest class. Red and orange **frames** are among the hottest. Know which part you're reading.

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CARD 1 OF 8 BULB COLORS **ANSWER**

Red or orange: Ordinary (135–170°F)
Yellow or green: Intermediate (175–225°F)
Blue: High (250–300°F)
Purple: Extra high (325–375°F)
Black: the three hottest classes (very extra high and both ultra high, 400°F and up)

NFPA 13 (2019) · Ch. 7

Trap: the color gives you the **class**, not the exact rating. A red bulb can be any rating from 135 to 170°F.

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CARD 4 OF 8 NO COLOR NEEDED **ANSWER**

No color needed: ornamental heads finished at the factory (plated or painted), and **recessed, flush and concealed** sprinklers.

Corrosion-resistant heads may show it with a **colored dot on the deflector**, a colored coating, or colored frame arms.

NFPA 13 (2019) · Ch. 7

Trap: "no color code" doesn't mean "no rating." Check the head's markings or submittal before you swap it.

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CARD 3 OF 8 BULB HEADS **ANSWER**

The liquid in the bulb. It must be colored to match the temperature table.

The frame arms of a bulb sprinkler don't have to be color coded at all.

So on a bulb head, read the bulb. A plain brass frame tells you nothing about its rating.

NFPA 13 (2019) · Ch. 7

On the job: a chrome pendent with a green bulb is intermediate, even though the frame has no color on it.

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sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 PICKING A RATING **QUESTION**

The ceiling over a process area runs about **140°F**. What temperature class do the sprinklers need?

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CARD 6 OF 8 HEAT SOURCES **QUESTION**

Which **heat sources** force a hotter sprinkler next to them? Name two that need intermediate and one that needs high.

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CARD 7 OF 8 HOT SPOTS **QUESTION**

Name **four locations** where NFPA 13 calls for intermediate-temperature heads (or hotter) even with no heater nearby.

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CARD 8 OF 8 STORAGE & SPARES **QUESTION**

In a **storage** occupancy, what's the lowest rating allowed for **ceiling** sprinklers? And what goes in the spare cabinet?

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*FSM · Sprinkler Fact Cards — NFPA 13 (2019) Temperature Ratings & Color Codes · sheet 2 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 6 OF 8 HEAT SOURCES **ANSWER**

Intermediate: heads within **12 in. to the side** of, or **30 in. above**, a bare steam main, heating coil or radiator.

High: heads within **7 ft** of a low-pressure blowoff valve that vents freely into a large room.

Commercial cooking: high or extra high, set by actually measuring the temperature.

Unit heaters: use the distance table and figure.

NFPA 13 (2019) · Ch. 9

On the job: a new branch line runs right over an old radiator. That head gets bumped to intermediate.

CARD 5 OF 8 PICKING A RATING **ANSWER**

Intermediate. Once the ceiling gets hotter than 100°F, you pick the class from the table's **maximum ceiling temperature** column:
Ordinary: up to 100°F
Intermediate: up to 150°F
High: up to 225°F
Extra high: up to 300°F

Default: ordinary and intermediate heads throughout, unless a rule says otherwise.

NFPA 13 (2019) · Ch. 7 · Ch. 9

Trap: comparing the ceiling temperature to the head's **rating**. You compare it to the **max ceiling** column.

CARD 8 OF 8 STORAGE & SPARES **ANSWER**

Storage ceiling heads: 150°F minimum (general, rack, rubber tire, roll paper and baled cotton storage). High-temperature heads may be used throughout storage, ordinary and extra hazard.

Spares must match the types and ratings installed. If the occupancy changes and so does the heat, the heads change too.

NFPA 13 (2019) · Ch. 9 · Ch. 16

On the job: a green-bulb head fused. Grabbing a red-bulb spare is wrong. Replace it with the same rating.

CARD 7 OF 8 HOT SPOTS **ANSWER**

Any four of:
• Glass or plastic skylights that get direct sun
• Unventilated attics and concealed spaces, or right under an uninsulated roof
• Unventilated show windows with bright lights up high
• Walk-in coolers and freezers with automatic defrost (intermediate **or higher**)
• Closets with ventless clothes dryers (intermediate **or higher**)

NFPA 13 (2019) · Ch. 9

Trap: the walk-in freezer. It's cold most of the day, but the defrost cycle heats the ceiling.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back